

Tabelle: Formeln der Mechanik und der Elektrotechnik (Beispiele)			
Formel	Umstellung 1	Umstellung 2	Umstellung 3
$W = F \cdot s$	$s = \frac{W}{F}$	$F = \frac{W}{s}$	
$F = m \cdot g \cdot h$	$m = \frac{F}{g \cdot h}$	$g = \frac{F}{m \cdot h}$	$h = \frac{F}{m \cdot g}$
$P = \frac{W}{t}$	$W = P \cdot t$	$t = \frac{W}{P}$	
$P = \frac{F \cdot s}{t}$	$F = \frac{P \cdot t}{s}$	$s = \frac{P \cdot t}{F}$	$t = \frac{F \cdot s}{P}$
$A = \frac{\pi}{4} \cdot d^2$	$d^2 = \frac{4 \cdot A}{\pi}$	$d = \sqrt{\frac{4 \cdot A}{\pi}} = 2 \sqrt{\frac{A}{\pi}}$	
$Q = n \cdot e$	$n = \frac{Q}{e}$	$e = \frac{Q}{n}$	
$U_{21} = \varphi_2 - \varphi_1$	$\varphi_2 = U_{21} + \varphi_1$	$\varphi_1 = \varphi_2 - U_{21}$	
$I = \frac{Q}{t}$	$t = \frac{Q}{I}$	$Q = I \cdot t$	
$I_1 + I_2 = I_3 + I_4$	$I_2 = I_3 + I_4 - I_1$	$I_4 = I_1 + I_2 - I_3$	$I_3 = I_1 + I_2 - I_4$
$R = \frac{l}{\gamma \cdot A}$	$A = \frac{l}{\gamma \cdot R}$	$\gamma = \frac{l}{R \cdot A}$	$l = R \cdot \gamma \cdot A$
$I = \frac{U}{R}$	$R = \frac{U}{I}$	$U = I \cdot R$	
$\frac{U_1}{U_2} = \frac{R_1}{R_2}$	$U_1 = U_2 \cdot \frac{R_1}{R_2}$	$U_2 = U_1 \cdot \frac{R_2}{R_1}$	$R_1 = R_2 \cdot \frac{U_1}{U_2}$
$P = U \cdot I$	$I = \frac{P}{U}$	$U = \frac{P}{I}$	
$P = \frac{U^2}{R}$	$U^2 = P \cdot R$	$U = \sqrt{P \cdot R}$	$R = \frac{U^2}{P}$
$W = U \cdot I \cdot t$	$I = \frac{W}{U \cdot t}$	$U = \frac{W}{I \cdot t}$	$t = \frac{W}{U \cdot I}$
$P = I^2 \cdot R$	$I^2 = \frac{P}{R}$	$I = \sqrt{\frac{P}{R}}$	$R = \frac{P}{I^2}$